7/24/2026

[Shadow Fox Internship]

[Beginner Level Task]

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[SHADOWFOX DATA SCIENCE INTERNSHIP]

Introduction Of Data Visualization

What is Data Visualization?

Data visualization is the process of representing data in graphical or visual forms such as charts, graphs, maps, and dashboards. It helps people understand complex by revealing patterns, trends, relations, and outliers that are difficult to identify from raw numbers alone.

Importance Of Data Visualization:

- Makes complex data easy to understand.
- Helps identify trends and patterns.
- Supports better decision-making.
- Detects outliers and anomalies.

Introduction Of Matplotlib

Matplotlib is one of the most popular and widely used data visualization libraries in Python. It was developed by John D. Hunter in 2002. It provides a flexible and powerful way to create high-quality 2D graphs and charts for data analysis and presentation.

Matplotlib allows users to create a wide variety of visualizations, including line plots, bar charts, scatter plots, histograms, pie charts, and many more.

Importance Of Matplotlib:

1. It helps convert raw data into meaningful visual representations.
2. It makes complex datasets easier to understand and analyse.
3. It is widely used in data science, machine learning, business analytics, and scientific research.

Installation:

(pip install matplotlib)

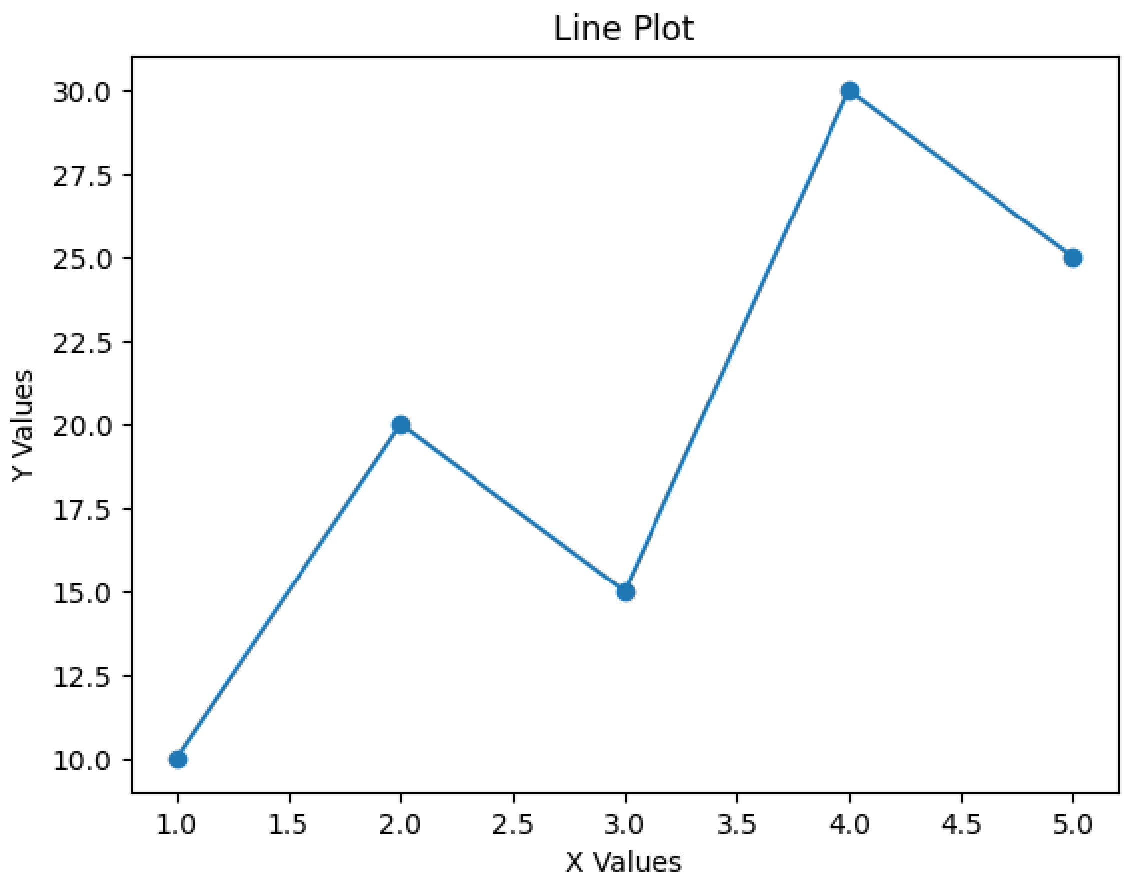
Some Graphs or Plots for Matplotlib:

1. # Line Plot

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,15,30,25]

plt.plot(x, y, marker='o')
plt.title("Line Plot")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.show()
```



Explanation Of Line Plot for Matplotlib:

A line plot is one of the most commonly used graphs in Matplotlib. It is used to show how data changes over a continuous period of time or in a sequence.

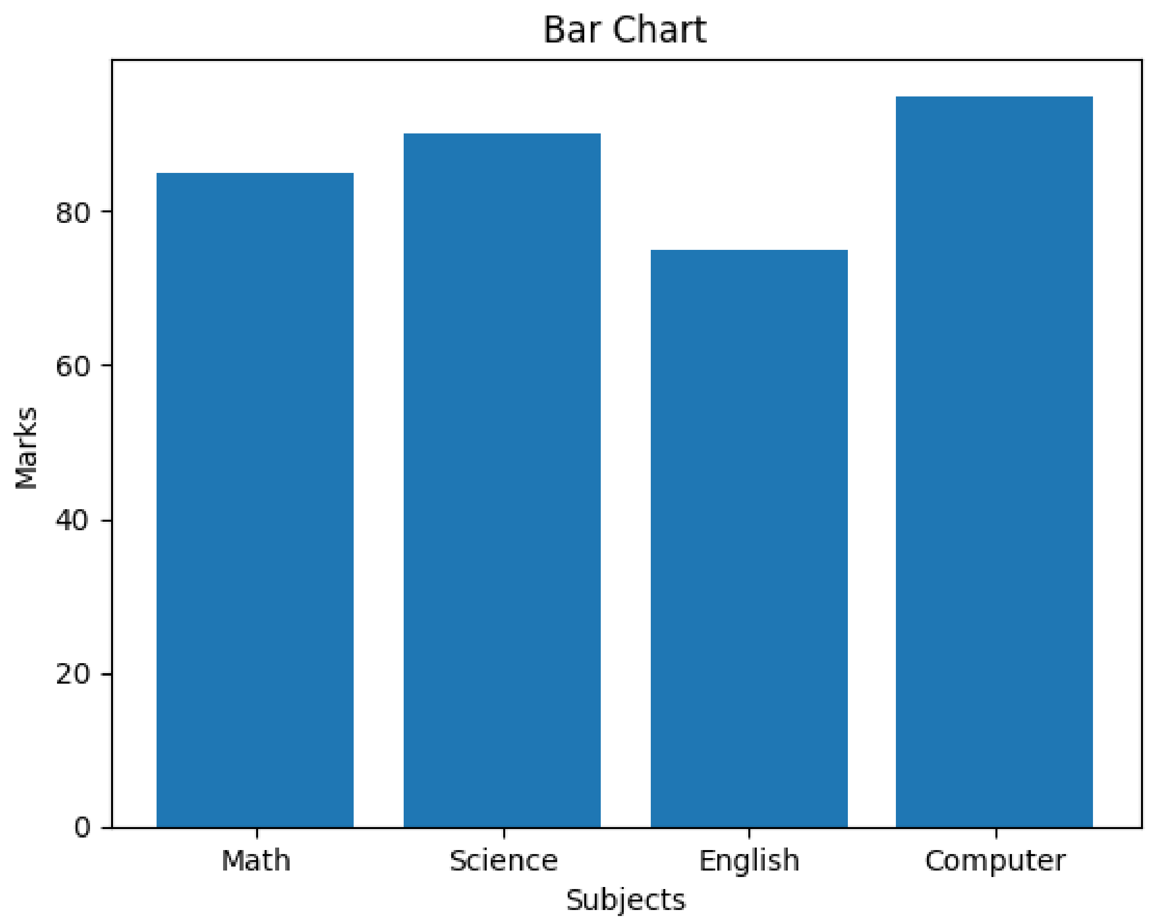
In a line plot, individual data points are connected by straight lines, which makes it easy to observe trends, growth, decline, and fluctuations in the data.

2. # Bar Chart

```
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English", "Computer"]
marks = [85, 90, 75, 95]

plt.bar(subjects, marks)
plt.title("Bar Chart")
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.show()
```



Explanation Of Bar Chart for Matplotlib:

A Bar graph is one of the most commonly used graphs in Matplotlib. It is used to compare different categories of data using rectangular bars. Bar charts make it easy to compare quantities and identify the highest and lowest values among different groups.

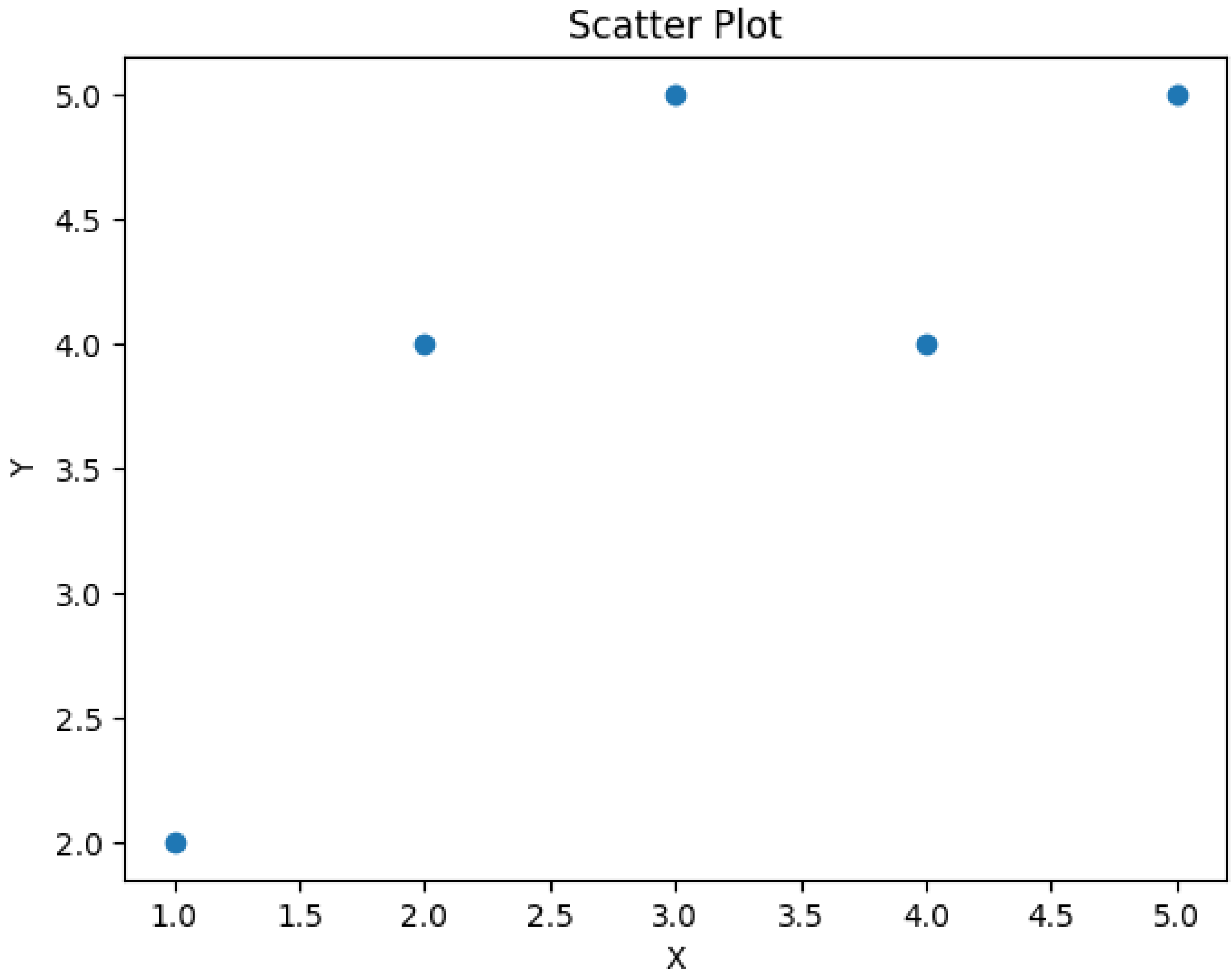
3. # Scatter Plot

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [2,4,5,4,5]

plt.scatter(x, y)
plt.title("Scatter Plot")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```

n



Explanation of Scatter Graph for Matplotlib:

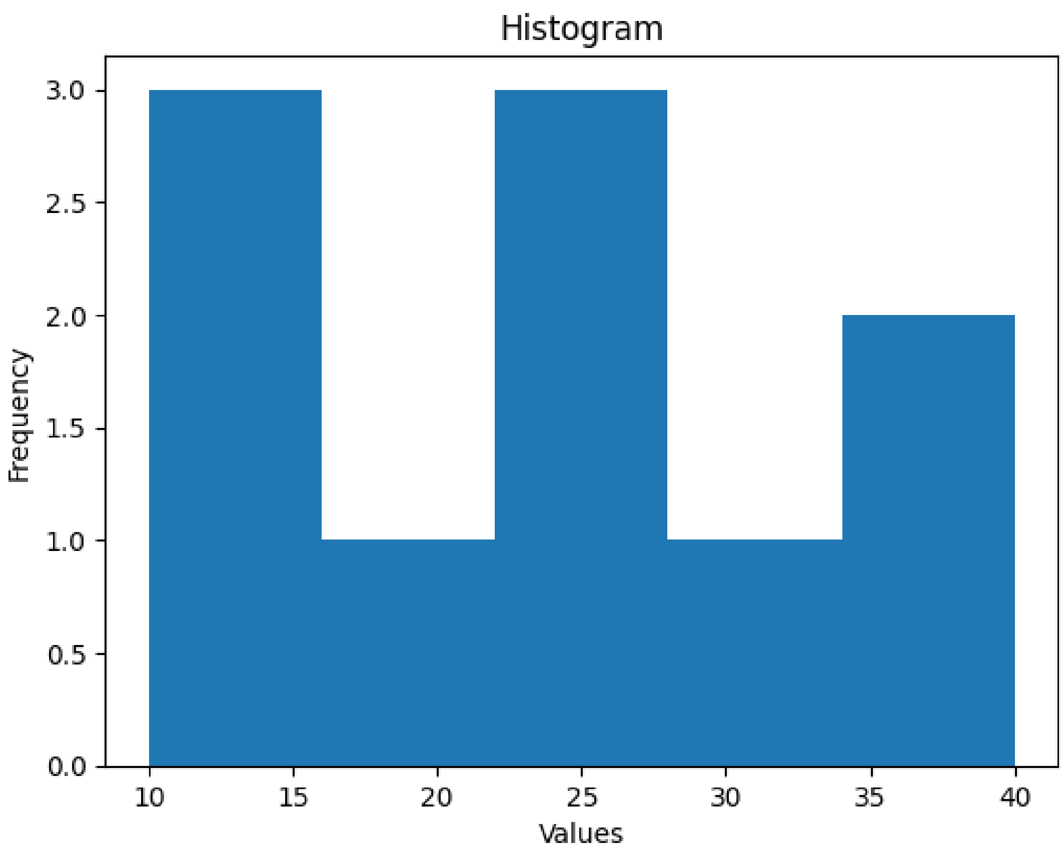
A Scatter Plot is a type of graph used to display the relationship between two numerical variables. It represents data as individual points on a coordinate plane, where each corresponds to one observation.

4. # Histogram

```
import matplotlib.pyplot as plt


data = [10,12,15,20,22,25,25,30,35,40]


plt.hist(data, bins=5)
plt.title("Histogram")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```



Explanation Of Histogram for Matplotlib:

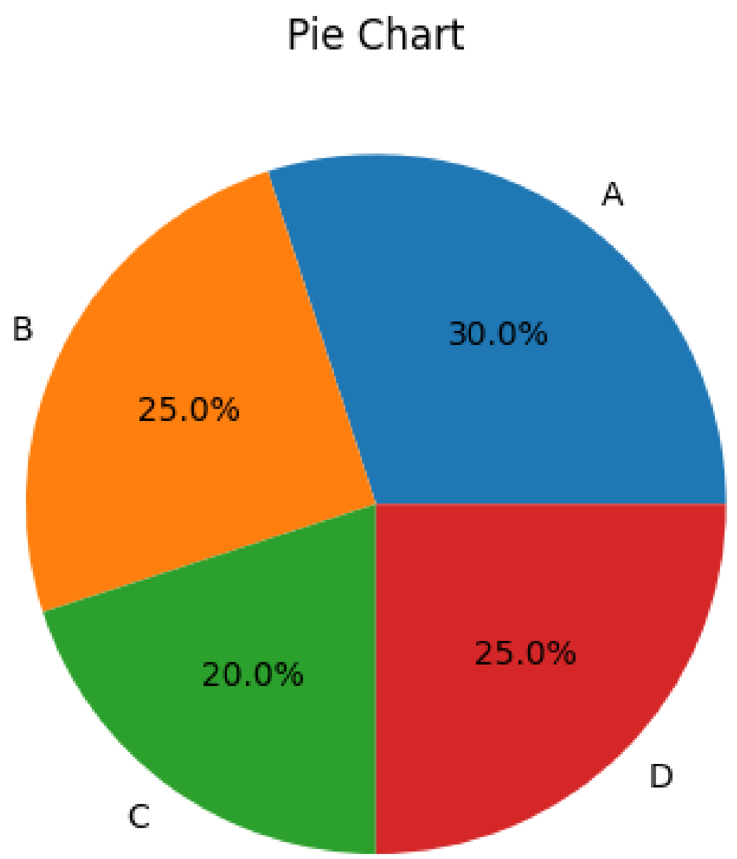
A Histogram is graph used to display the distribution of numerical data. It divides the data into a series of continuous intervals called bins and shows the frequency of data points that fall within each interval. It is used for continuous data.

5. # Pie Chart

```
import matplotlib.pyplot as plt

labels = ["A", "B", "C", "D"]
sizes = [30,25,20,25]

plt.pie(sizes, labels=labels, autopct="%1.1f%%")
plt.title("Pie Chart")
plt.show()
```



Explanation of Pie Chart for Matplotlib:

A Pie Chart is a circular statistical graph used to represent data as proportions or percentage of a whole. The chart is divided into slices, where each slice represents a category, and the size of each slice is proportional to its value. Pie charts are most effective when showing how different categories contribute to a total.

Introduction Of Pandas-

Pandas is one of the most powerful and widely used open-source Python libraries for data manipulation and data analysis. It was developed by Wes Mckinney in 2008. Pandas is commonly used in data science, machine learning, business analytic, finance, and research.

Importance Of Pandas:

1. It helps organize and analyse large datasets efficiently.
2. It provides powerful data structures like Series and Data Frame.
3. It makes data cleaning and preprocessing simple.
4. It supports importing such as CSV, Excel, JSON, and SQL.

Installation:

(pip install pandas)

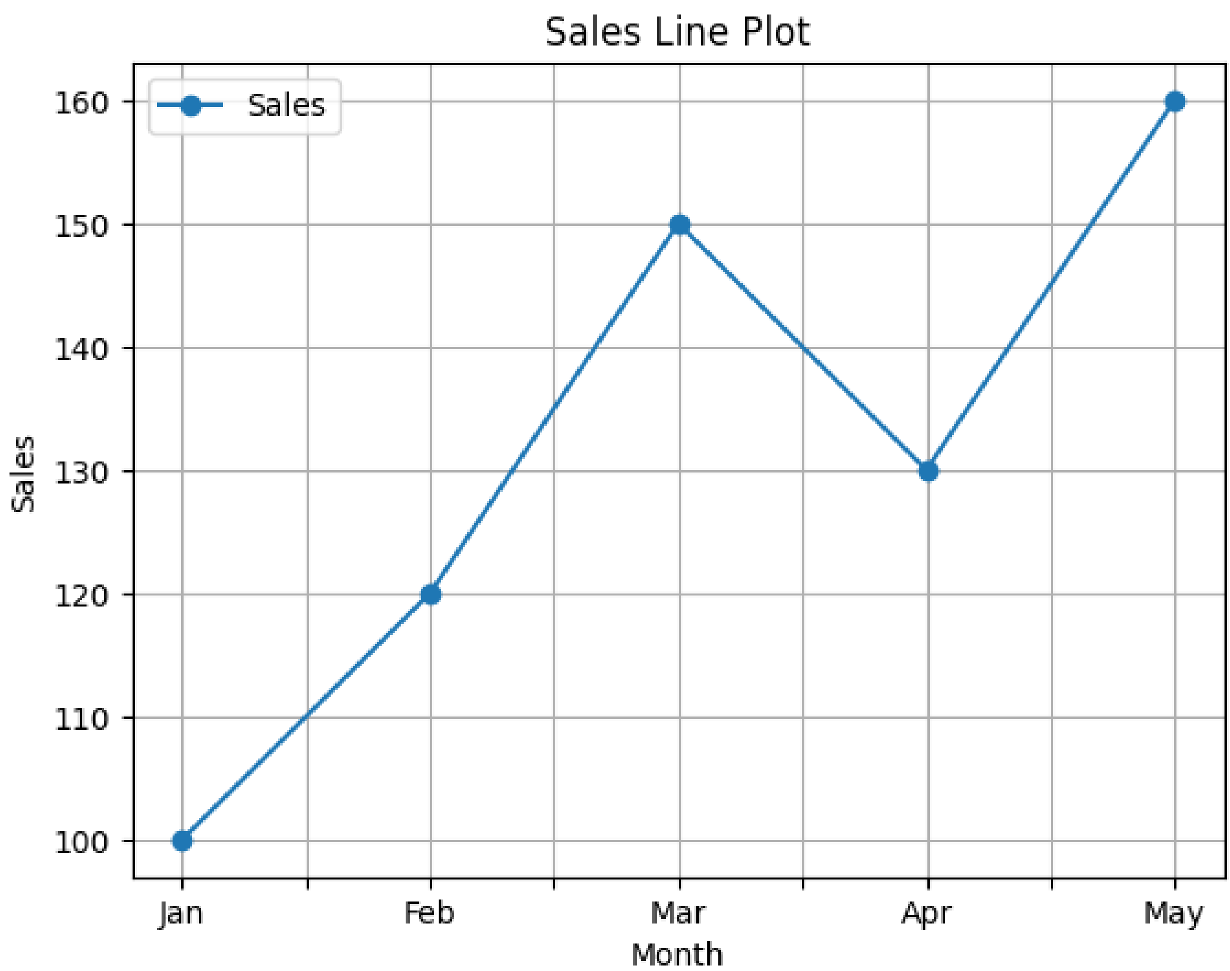
Some Graphs/Plots for Pandas:

1. # Line Plot

```
import matplotlib.pyplot as plt
import pandas as pd

# Sample data for demonstration
data = {
  "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
  "Sales": [100, 120, 150, 130, 160]
}
df = pd.DataFrame(data)

df.plot(x="Month", y="Sales", kind="line", marker='o')
plt.title("Sales Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



Explanation of Line Plot for Pandas:

A line plot is one of the most commonly used visualization techniques in Pandas. It is used to display changes or trends in data over a continuous period, such as time, dates, or sequential values. In a line plot, individual data points are connected by straight lines.

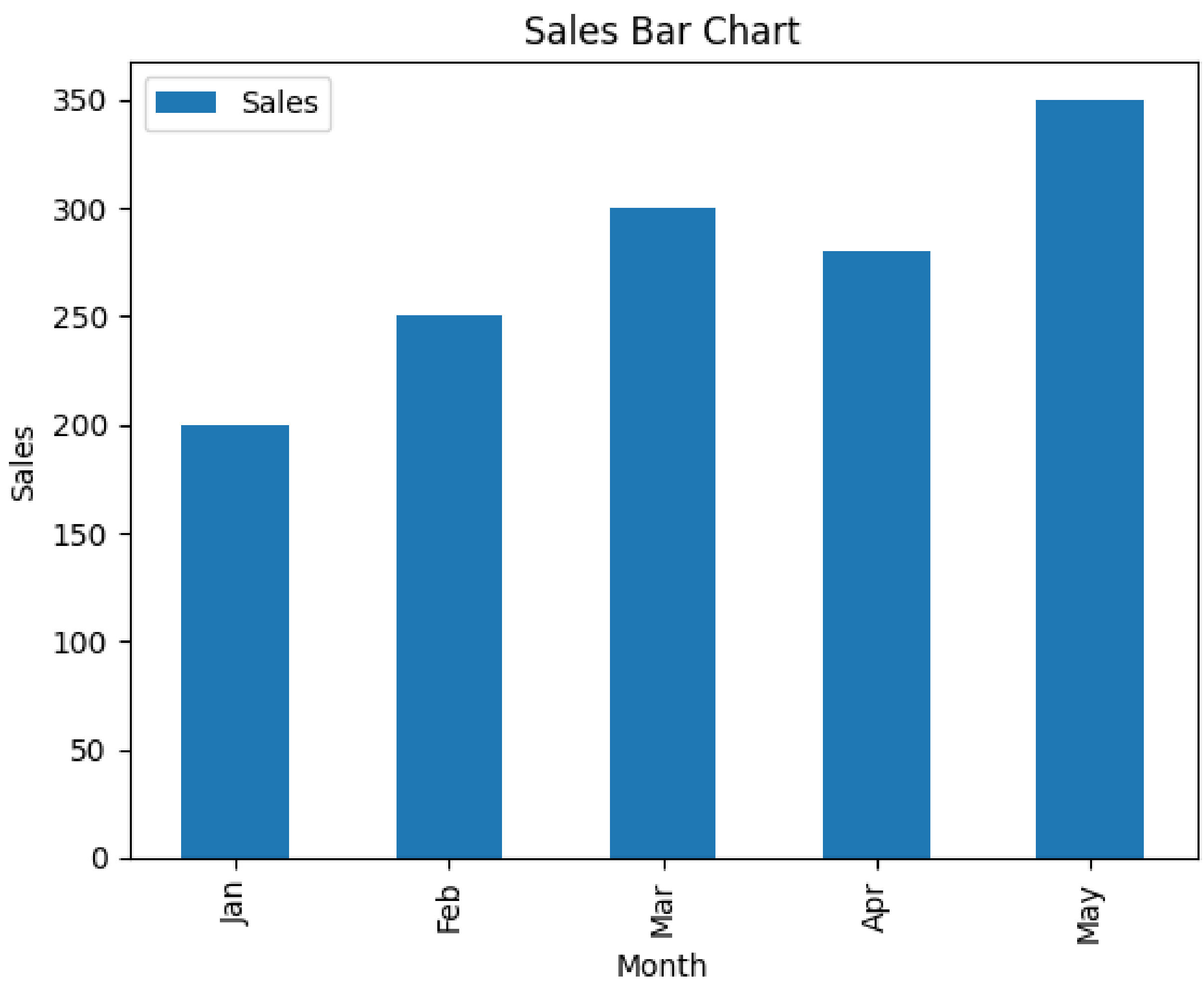
2. # Bar Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Create sample DataFrame
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [200, 250, 300, 280, 350]
}

df = pd.DataFrame(data)

# Bar Chart
df.plot(x="Month", y="Sales", kind="bar")
plt.title("Sales Bar Chart")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```



Explanation of Bar Chart for Pandas:

A Bar Chart is a graphical representation used to compare values across different categories. Pandas uses Matplotlib as the default visualization library to generate bar charts. They provide a clear and simple way to identify the highest and lowest values among categories.

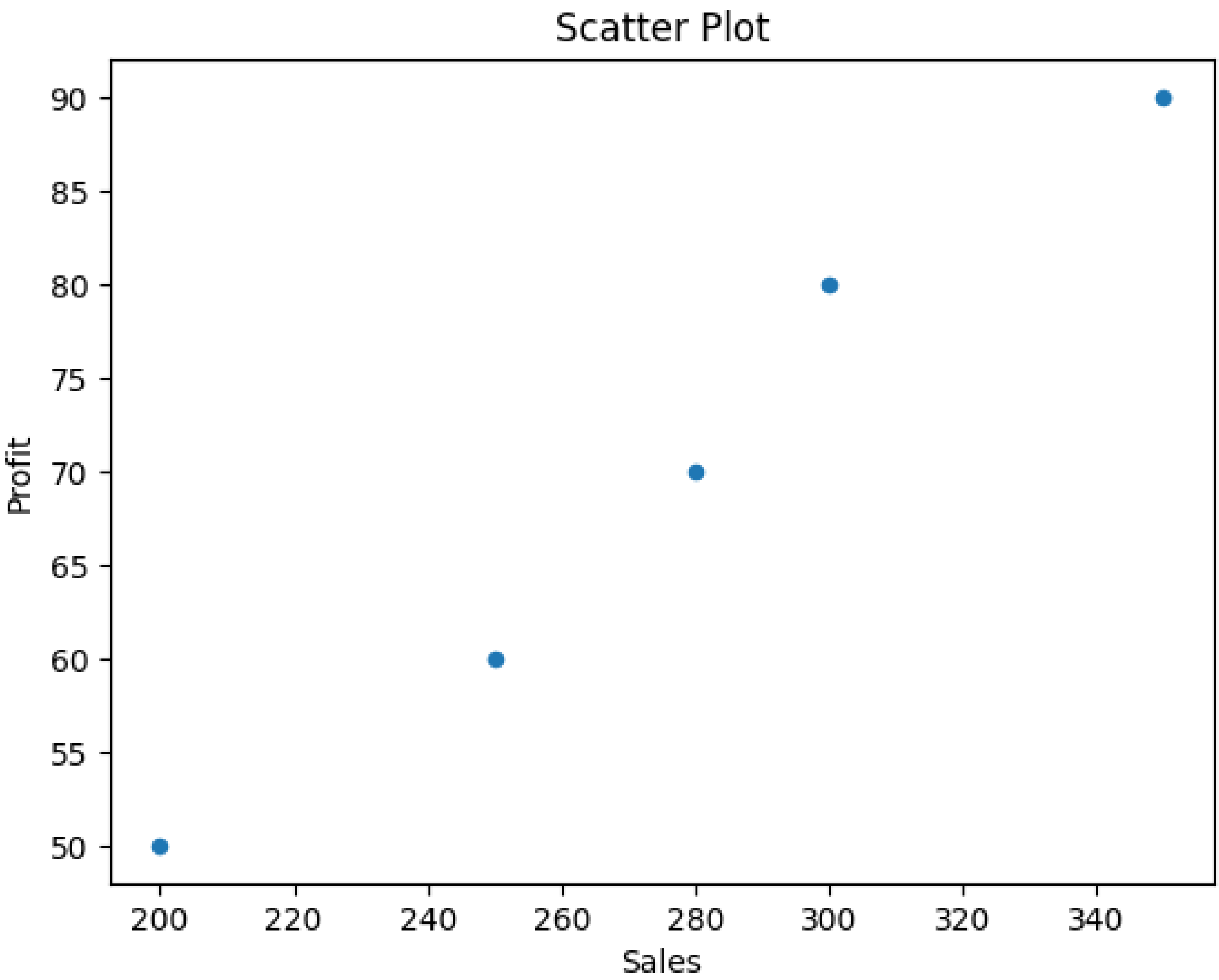
3. # Scatter Plot

```
import pandas as pd
import matplotlib.pyplot as plt

# Create sample DataFrame
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [200, 250, 300, 280, 350],
    "Profit": [50, 60, 80, 70, 90]
}

df = pd.DataFrame(data)

# Scatter Plot
df.plot(x="Sales", y="Profit", kind="scatter")
plt.title("Scatter Plot")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()
```



Explanation of Scatter Plot for Pandas:

A Scatter Plot is a graphical representation used to display the relationship between two numerical variables. Scatter plots help identify patterns, trends, correlations, clusters, and outliers in a dataset. Pandas uses Matplotlib as its default visualization backend, making it easy to create scatter plots with minimal code.

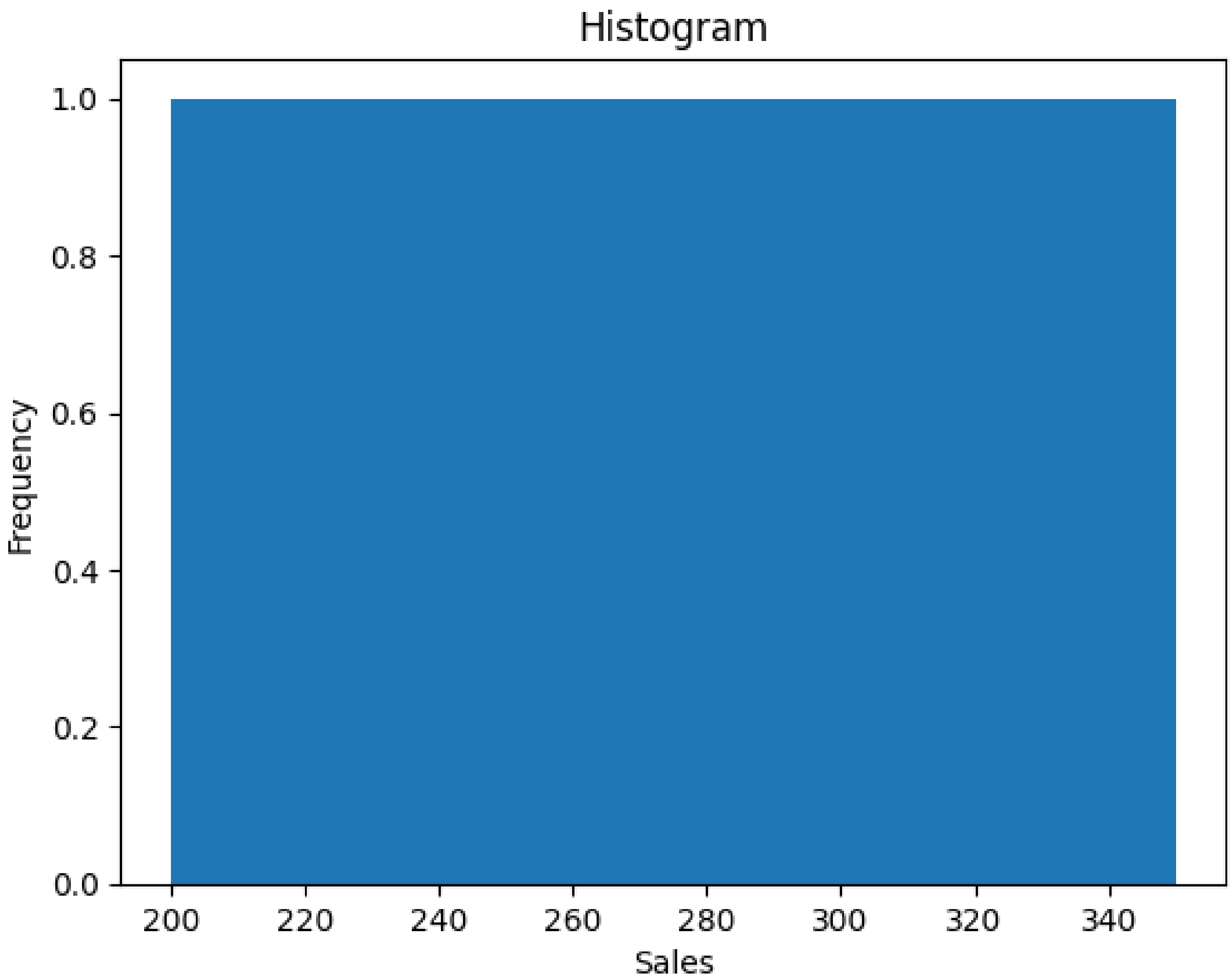
4. #Histogram

```
import pandas as pd
import matplotlib.pyplot as plt

# Create sample DataFrame
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [200, 250, 300, 280, 350],
    "Profit": [50, 60, 80, 70, 90]
}

df = pd.DataFrame(data)

# Histogram
df["Sales"].plot(kind="hist", bins=5)
plt.title("Histogram")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```



Explanation of Histogram for Pandas:

A Histogram is a graphical representation used to display the frequency distribution of numerical data. Unlike a bar chart, the bars in a histogram are adjacent because the data is continuous.

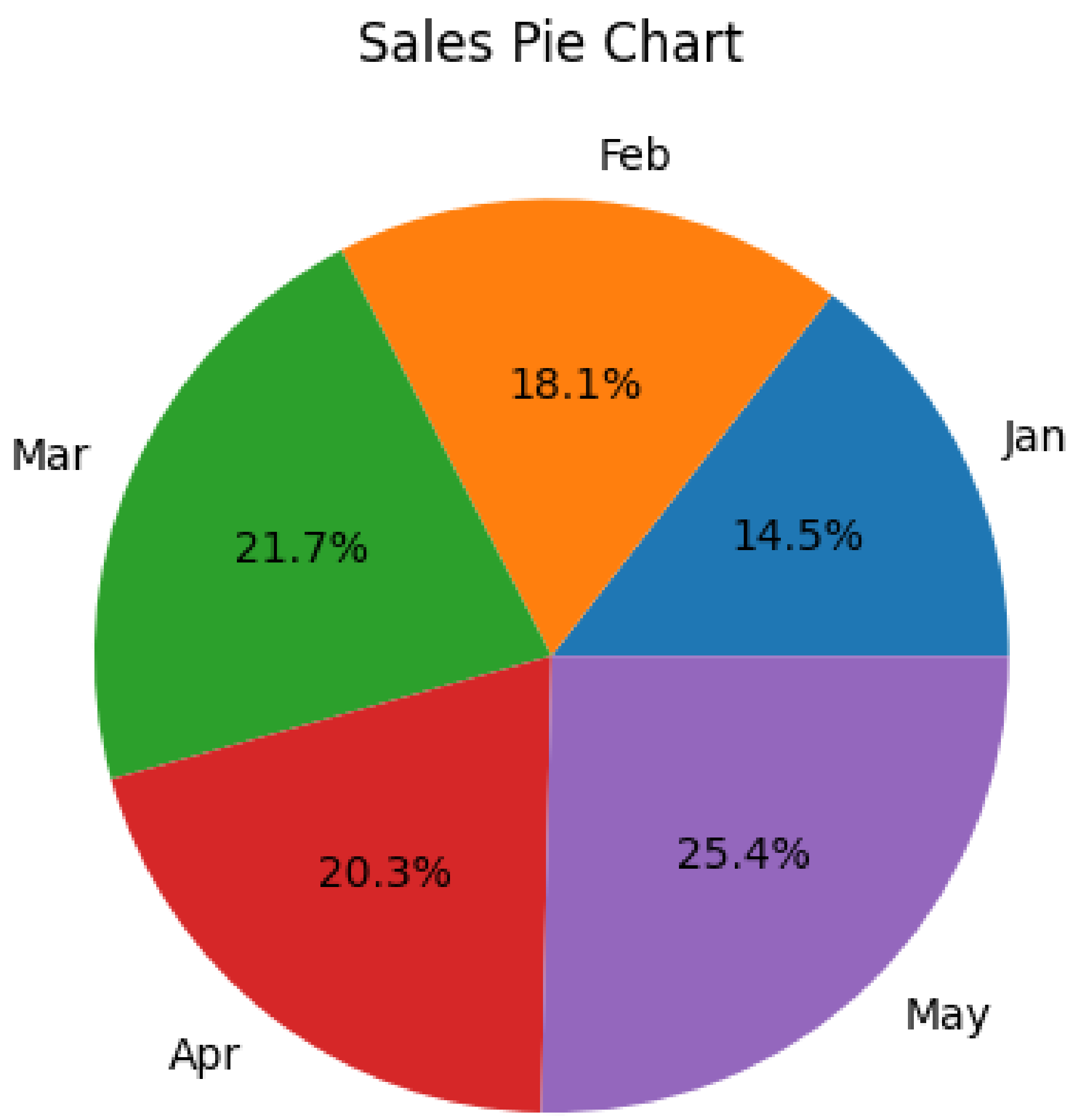
5. #Pie Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Create sample DataFrame
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [200, 250, 300, 280, 350]
}

df = pd.DataFrame(data)

# Pie Chart
df.set_index("Month")["Sales"].plot(kind="pie",
autopct="%1.1f%%")
plt.title("Sales Pie Chart")
plt.ylabel("")
plt.show()
```



Explanation of Pie Chart for Pandas:

A Pie Chart is a circular graphical representation used to display the proportion or percentage of different categories in a dataset. Pie charts are best suited for showing how individual categories contribute to a whole.

Comparison of Matplotlib and Pandas:

Matplotlib	Pandas
Matplotlib is a dedicated Python library used for creating a wide variety of graphs and charts.	Pandas is Python library mainly used for data manipulation, cleaning, and analysis.
It is ideal for creating highly customized and professional-quality visualizations.	It is ideal for quick and easy visualizations of data stored in Data frame and Series.
It requires more code to create and customize graphs.	It requires less code because plotting functions are integrated with Data Frame.
Low – level library.	High – level library.
More flexible for Beginners.	Easy for Beginners.

Conclusion:

Both Matplotlib and Pandas are essential libraries in Python for data analysis and visualization. Matplotlib is best suited for creating detailed and highly customizable charts, while Pandas is ideal for quick data analysis and simple visualization directly from Data Frames.